# GATE CS — Computer Networks

Concise study notes for GATE 2027 · built from the 2016–2026 papers in this repo

## Where the marks come from (average marks per paper, 2016–2026)

| Topic | Marks / paper | Priority |
|---|---|---|
| IP: addressing, CIDR, fragmentation, NAT | 2.2 | Tier 2 |
| Data link: MAC, error detection, sliding window | 2.1 | Tier 2 |
| Application layer, layering, DNS/HTTP, misc | 1.9 | Tier 2 |
| TCP: flow & congestion control | 1.5 | Tier 3 |
| Routing (DV/LS) | 0.4 | Tier 3 |

Computer Networks carries about **8.1 of the 85 technical marks** in a typical paper. Topics below are listed in the order given in the notes; use the table above to decide where to spend time first.

Legend: **Trap** = common mistake · **Tip** = shortcut. For practice, use `GATE_CS_Computer_Networks.pdf` (past questions + new questions with solutions). These notes were written from general knowledge, not checked line by line against a textbook; verify the less common facts.

# 1. IP Addressing, Subnetting, CIDR, Fragmentation, NAT

## IPv4 addressing

- 32-bit address; classes: A 0.0.0.0–127.255.255.255 (/8), B 128–191 (/16), C 192–223 (/24), D 224–239 (multicast), E 240–255. Private ranges: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16. Loopback 127.0.0.0/8. Limited broadcast 255.255.255.255; 0.0.0.0 = “this host”.
- In a subnet with h host bits: $2^h$ addresses, **$2^h - 2$ usable hosts** (network and broadcast addresses excluded). Network address = IP AND mask. Broadcast = network OR (NOT mask).
- Mask /n → n ones. 255.255.255.192 = /26 (6 host bits → 62 hosts). Quick masks: 128 = /1 in the octet, 192 = /2, 224 = /3, 240 = /4, 248 = /5, 252 = /6, 254 = /7.
- **Subnetting**: borrow bits from the host part. To split into k subnets, borrow $\lceil \log_2 k \rceil$ bits. **VLSM**: allocate the largest subnets first.
- **CIDR**: a.b.c.d/n; **route aggregation** (supernetting) merges contiguous blocks aligned on a power of two. A block of size $2^k$ must start at an address divisible by $2^k$.
- **Longest-prefix match:** when several forwarding-table entries match, use the one with the longest mask. A default route 0.0.0.0/0 matches everything with the lowest priority.
- Forwarding: the router ANDs the destination with each entry's mask and compares with the entry's network.

## IP datagram and fragmentation

- IPv4 header: 20 bytes minimum (up to 60); fields: version, IHL, total length (16 bits, max 65,535), identification, flags (DF, MF), fragment offset (13 bits, in units of 8 bytes), TTL, protocol, header checksum, source, destination.
- Fragmentation: payload per fragment = MTU − header, rounded down to a multiple of 8 (except the last). Offset of fragment k = (bytes before it)/8. MF = 1 on all but the last. Reassembly happens only at the destination. If DF is set and the packet exceeds the MTU, the router drops it and sends ICMP.
- TTL decrements at each router; at 0 the packet is dropped (ICMP time exceeded) — also used for traceroute. The header checksum is recomputed at each hop.
- **IPv6:** 128-bit addresses, fixed 40-byte header, no router fragmentation, no header checksum. **NAT**: translates private to public addresses (and ports); breaks end-to-end addressing. **ICMP** reports errors (ping uses echo). **ARP** maps IP → MAC (broadcast request, unicast reply); **DHCP** assigns addresses dynamically (DORA: discover, offer, request, ack).

# 2. Data Link Layer: Error Detection, Sliding Window, MAC

## Error detection and correction

- **Parity:** detects any odd number of bit errors. **CRC:** divide the data (shifted left by the degree of the generator) by the generator polynomial using XOR; the remainder is appended. Receiver divides again; remainder 0 = no error detected. A generator of degree r detects all burst errors of length ≤ r, all single-bit and all odd-number errors if x+1 is a factor.
- **Checksum** (Internet checksum): ones' complement sum of 16-bit words, then complemented.
- **Hamming code:** for m data bits and r check bits, $2^r \geq m + r + 1$. Check bits at positions 1, 2, 4, 8… Minimum Hamming distance d: detects $d - 1$ errors and corrects $\lfloor (d - 1)/2 \rfloor$.

## Flow control and sliding windows

- Transmission time T_t = frame size / bandwidth; propagation delay T_p = distance / speed. Round trip = 2T_p.
- **Stop-and-wait** efficiency = T_t / (T_t + 2T_p) (ignoring ACK time). Utilisation with window size N (sliding window) = N·T_t / (T_t + 2T_p), capped at 1. Minimum window to keep the link busy: N = 1 + 2T_p/T_t.
- **Go-Back-N:** sender window N ($N \leq 2^k - 1$ for k-bit sequence numbers), receiver window 1; a lost frame makes the sender resend the frame and everything after it. **Selective Repeat:** both windows $N \leq 2^{k-1}$; only the lost frame is resent. Cumulative ACKs in GBN.
- Bits for sequence numbers: GBN needs $\lceil \log_2(N + 1) \rceil$; SR needs $\lceil \log_2(2N) \rceil$.
- Piggybacking: the ACK rides on a data frame going the other way.

## Medium access and Ethernet

- **Pure ALOHA:** max efficiency 1/(2e) ≈ 18.4% (at G = 0.5); **slotted ALOHA** 1/e ≈ 36.8% (at G = 1). Throughput S = G·e^(−2G) (pure), G·e^(−G) (slotted).
- **CSMA/CD** (Ethernet): listen before sending, detect collisions while transmitting, exponential backoff. For collision detection, the frame transmission time must be ≥ 2T_p: minimum frame size = 2·T_p·bandwidth. Classic Ethernet minimum frame 64 bytes; MTU 1500 bytes; MAC address 48 bits.
- Bridges/switches learn MAC addresses and filter at the data link layer; each switch port is its own collision domain; a hub repeats at the physical layer (one collision domain). Wi-Fi uses CSMA/CA.
- Framing methods: character count, byte stuffing, bit stuffing (insert 0 after five consecutive 1s).

## 3. Application Layer, Layering, Delay

- **OSI (7 layers):** physical, data link, network, transport, session, presentation, application. **TCP/IP (4–5 layers):** link, internet, transport, application. Each layer adds a header (encapsulation).
- Device layers: hub/repeater – physical; bridge/switch – data link; router – network; gateway – application. Protocols: HTTP 80 (TCP), HTTPS 443, FTP 20/21, SSH 22, Telnet 23, SMTP 25, DNS 53 (UDP for queries, TCP for zone transfers), DHCP 67/68 (UDP), POP3 110, IMAP 143, SNMP 161.
- **DNS:** hierarchical distributed database; resolvers query root → TLD → authoritative servers; iterative vs recursive queries; caching with TTL. Record types: A, AAAA, MX, NS, CNAME, PTR.
- **HTTP:** stateless request/response; methods GET, POST, PUT, DELETE, HEAD; HTTP/1.1 uses persistent connections (and pipelining); cookies add state; status codes 200, 301, 404, 500. Non-persistent HTTP takes 2 RTT per object (connection + request); persistent without pipelining 1 RTT per object. **Email:** SMTP pushes, POP3/IMAP pulls. **FTP** uses separate control and data connections.
- **Delay:** total = processing + queueing + transmission (L/R) + propagation (d/s). Store-and-forward over k links with packet size L: end-to-end delay = k·L/R (ignoring propagation). For a message of N packets of size L over k links: $(N + k - 1)\cdot L/R$. Bandwidth-delay product = R × RTT (bits in flight).
- Circuit switching (reserved path, no queueing) vs packet switching (statistical multiplexing, queueing). Number of users supported by a link with circuit switching = R/rate.
- Network security basics: symmetric (shared key, AES, DES) vs asymmetric (RSA, public/private key) encryption; digital signatures; hash functions (SHA, MD5); firewalls; IPsec, SSL/TLS.

## 4. TCP: Flow and Congestion Control

- TCP: reliable, connection-oriented byte stream. Header 20–60 bytes; 16-bit ports; sequence number counts bytes; ACK number = next byte expected. **3-way handshake**: SYN, SYN+ACK, ACK; **teardown**: FIN, ACK, FIN, ACK (4 segments). TIME_WAIT lasts 2×MSL.
- **Flow control:** receiver advertises a window (rwnd). **Congestion control:** sender keeps cwnd; usable window = min(cwnd, rwnd). Throughput ≈ window / RTT. Link utilisation = (W·MSS/RTT)/bandwidth.
- **Slow start:** cwnd = 1 MSS, doubles every RTT until it reaches ssthresh. **Congestion avoidance (AIMD):** cwnd += 1 MSS per RTT. On a **timeout:** ssthresh = cwnd/2, cwnd = 1 MSS, restart slow start. On **3 duplicate ACKs (fast retransmit/recovery, Reno):** ssthresh = cwnd/2, cwnd = ssthresh (+3), continue in congestion avoidance.
- Example: ssthresh 16, start cwnd 1: rounds 1–5 → 1, 2, 4, 8, 16; then 17, 18, 19 …
- RTT estimation: EstimatedRTT = (1−α)·EstimatedRTT + α·SampleRTT (α = 1/8); timeout = EstimatedRTT + 4·DevRTT. Karn's algorithm: don't sample retransmitted segments.
- Maximum window without wrap-around: sequence space is $2^{32}$ bytes, so wrap-around time = $2^{32}$/bandwidth. **Silly window syndrome**, Nagle's algorithm, delayed ACKs.
- **UDP:** connectionless, unreliable, 8-byte header (ports, length, checksum); used for DNS, streaming, DHCP. TCP's MSS = MTU − 40 (IP + TCP headers, no options).
- Typical numerical question: given bandwidth, RTT and window, find utilisation or throughput; the number of bits for a window of W bytes; time to send a file with slow start (count RTTs).

## 5. Routing: Distance Vector and Link State

- **Distance vector (Bellman–Ford, RIP):** each router sends its vector of costs to its neighbours; D_x(y) = min over neighbours v of [c(x,v) + D_v(y)]. Converges iteratively; suffers from **count-to-infinity** (bad news travels slowly). Fixes: split horizon, poison reverse, a small infinity (RIP uses 16 hops). RIP: hop count metric, UDP, updates every 30 s.
- **Link state (Dijkstra, OSPF):** every router floods link-state advertisements, builds the full topology and runs Dijkstra. Faster convergence, more memory and bandwidth for flooding. OSPF runs directly over IP and supports areas.
- **BGP:** path-vector routing between autonomous systems (inter-domain), policy-based; runs over TCP port 179. RIP/OSPF are intra-AS (interior gateway protocols).
- Routing tables are computed in advance; forwarding is the per-packet lookup. **Static** vs **dynamic** routing; **flooding**, hot-potato routing.
- Running Dijkstra for the numerical question: keep a table of tentative distances; at each step finalise the closest vertex and relax its neighbours; sum distances as asked.
- Number of routers whose table changes after a link failure, or the number of iterations for a distance-vector update to converge, are common exam variants: simulate round by round.

## 6. Exam strategy & revision checklist

- Study order: IP addressing and subnetting → TCP/congestion control → data link (sliding window, CRC) → routing → application layer.
- Always convert units (bits vs bytes, ms vs s, KB = 1024 vs 1000 as stated).
- ☐ Subnet masks and hosts per subnet. ☐ Longest-prefix match. ☐ Fragmentation offsets. ☐ Window efficiency formulas. ☐ CRC division. ☐ Slow-start/AIMD trace. ☐ Distance-vector rounds. ☐ Redo `GATE_CS_Computer_Networks.pdf` past questions.